SHUSHRUTHA THANGALLAPALLY

Software Engineer | Cloud & AI/ML | Full-Stack Developer

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PROFESSIONAL SUMMARY

Final-year Computer Science (AI & ML) student with hands-on experience across Software Development, Machine Learning, Full-Stack Web Development, and Computer Vision. Skilled in building scalable web applications, AI-powered solutions, RESTful APIs, and cloud-based systems using Python, Java, JavaScript, React.js, Node.js, TensorFlow, and PyTorch. ServiceNow Certified System Administrator (CSA) and Certified Application Developer (CAD) with strong Data Structures & Algorithms, leadership, and software engineering skills. Seeking Software Engineer, AI Engineer, Machine Learning Engineer, or Full Stack Developer opportunities.

EDUCATION

Bachelor of Technology (B.Tech.), Computer Science and Engineering (AI & ML) | Sep 2023 – May 2027 (Expected)
CMR Technical Campus, Hyderabad — CGPA: 9.1/10.0

Foundation Level Certificate in Data Science and Applications | Completed Foundational Level : May 2026
IIT Madras (Remote)

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, JavaScript, C#.NET, Kotlin, SQL

Frameworks & Libraries: React.js, Express.js, Node.js, Angular, Flutter, Spring Boot, TensorFlow, PyTorch, OpenCV

Databases, Cloud & Platforms: MySQL, MongoDB, Firebase, AWS EC2, ServiceNow

Web Development: HTML5, CSS3, Bootstrap, jQuery, REST APIs, JSON, AJAX

Developer Tools: Git, GitHub, VS Code, Android Studio, Visual Studio, Postman, phpMyAdmin

Core Concepts: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Software Engineering, Machine Learning, SDLC, Agile, Generative AI, Agentic AI, RAG Applications

TECHNICAL PROJECTS

Agentic AI-Based Radiology Assistant: Disease Detection in Chest X-Rays with LLM Reasoning and RAG Integration

- Developed RadiologyAI, a multi-agent system using pretrained DenseNet-121 to analyze chest X-rays and identify 18 thoracic findings.
- Implemented a Gate-Analyse-Report (GAR) pipeline integrating image validation, deep learning-based classification, and LLM-powered structured report generation.
- Achieved 0.945 AUROC for Lung Opacity and improved Pneumonia AUROC from 0.582 to 0.791 through optimized image preprocessing on 624 pediatric X-rays.
- Integrated RAG-based medical knowledge retrieval, achieving 70% Hit@1 and 80% Hit@4 across 20 test questions using five medical reference notes.

SketchAI – AI-Enhanced Collaborative Drawing Game Platform – SketchAI

- Developed an interactive AI-powered drawing and guessing game integrating Machine Learning and Generative AI to enhance user engagement.
- Built a real-time multiplayer experience using Python Flask and Socket.IO, enabling synchronized gameplay and collaborative interactions.
- Integrated AI-driven sketch recognition and intelligent game features to enhance drawing interpretation and guessing experiences.
- Implemented backend services and database integration using MongoDB Atlas, with cloud deployment of the Flask backend on Render.

Smart Player Performance Analysis System (Kho-Kho) — *PlayVision*

- Built a real-time athlete tracking system using YOLOv8, OpenCV, and Python for sports analytics.
- Implemented object detection, player tracking, trajectory analysis, and movement analytics.
- Extracted performance metrics including reaction time, speed estimation, player positioning, and spatial analysis; optimized the inference pipeline for real-time video processing.

Institutional Web Infrastructure Development — *TGSWREIS Government Degree Colleges*

- Developed scalable PHP-MySQL web applications using MVC architecture.
- Designed responsive Bootstrap interfaces and secure authentication systems.
- Built CRUD modules, admin dashboards, CMS functionality, and reporting dashboards.
- Optimized SQL queries to improve application performance.

CERTIFICATIONS

- PwC Advisory Launchpad – CEDA - Cloud Engineering Track
- ServiceNow Certified System Administrator (CSA)
- ServiceNow Certified Application Developer (CAD)
- AI/ML Virtual Internship - Google for Developers
- NPTEL Certifications: Machine Learning, Cloud Computing, Data Mining, Blockchain and Its Applications, The Joy of Computing with Python, Internet of Things, Sensors and Actuators

ACHIEVEMENTS & EXTRACURRICULARS

- Led cross-functional teams in Smart India Hackathon (2024, 2025 Internal Rounds) and the NeuraX 24-Hour Hackathon, delivering AI-driven software solutions under time constraints.
- Active NSS volunteer contributing to community outreach initiatives while developing leadership, teamwork, and project coordination skills.
- Solved 100+ coding problems on LeetCode, strengthening proficiency in Data Structures, Algorithms, and problem-solving.
- Developed 8+ academic and real-world software projects spanning AI/ML, full-stack web development, and mobile applications.